

Third Semester B. Tech (CBCS) Degree Examinations

(IS)

Max. Marks: 100

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

14. The device which is allowed to initiate data transfers on the BUS at any time is called ____
- A) BUS master B) Processor
C) BUS arbitrator D) Controller
15. If during the execution of an instruction an exception is raised then ____
- A) The instruction is executed and the exception is handled B) The instruction is halted and the exception is handled
C) The processor completes the execution and saves the data and then handle the exception D) Exit from the current executed program and never return
16. The high speed mode of operation of the USB was introduced by ____
- A) ISA B) USB 3.0
C) USB 2.0 D) ANSI
17. The USB device follows ____ structure.
- A) List B) Huffman
C) Hash D) Tree
18. The number successful accesses to memory stated as a fraction is called as ____
- A) Access rate B) Success rate
C) Hit rate D) Miss rate
19. The drawback of building a large memory with DRAM is ____
- A) The Slow speed of operation B) The large cost factor
C) The inefficient memory organization D) All of the mentioned
20. The flash memory modules designed to replace the functioning of a hard disk is ____
- A) RIMM B) FIMM
C) Flash drives D) DIMM

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|---|-------|-----|-----|-----|
| 1. a) Explain the basic operational concepts of a computer system. Also draw a diagram of the different registers inside the processor. | 8 | L*2 | 1 | 1 |
| b) Derive basic performance equation, how can we achieve high performance. | 8 | L2 | 1 | 1 |
| 2. a) Explain condition codes specify commonly used flags. | 8 | L2 | 1 | 1 |
| b) What is sequential Multiplication? Explain with necessary diagram and examples. | 8 | L3 | 1 | 1 |
| 3. a) Briefly explain Booth Algorithm taking suitable example. | 8 | L3 | 2 | 2 |
| b) State the algorithm to perform restoring division. Solve using restoring division, M=00011(divisor), Q=1000(dividend). | 8 | L3 | 2 | 2 |

Unit – II

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|---|---|----|---|---|
| 4. a) Explain three bus organization with a neat diagram. Write the control sequence for the instruction Add R4, R5, R6 for the three bus organization. | 8 | L2 | 3 | 3 |
| b) Consider the instruction: ADD (R3),R1 Explain in detail about execution of this complete instruction. | 8 | L2 | 3 | 2 |
| 5. a) Explain the sequence of events in handling an Interrupt. | 8 | L2 | 4 | 2 |
| b) Explain with diagrams about Interrupt Priority schemes. | 8 | L2 | 4 | 2 |
| 6. a) With a neat diagram explain DMA controller in a computer system. | 8 | L2 | 4 | 2 |
| b) Provide a detailed explanation of what USB is, how it works. | 8 | L2 | 4 | 2 |
| 7. a) Explain direct mapping function in cache with neat diagram. | 8 | L3 | 5 | 2 |
| b) Explain how a Static RAM (SRAM) cell and CMOS cell can be implemented with a neat diagram. | 8 | L2 | 5 | 2 |

Unit – III

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|---|---|----|---|---|
| 8. a) Draw memory hierarchy considering Speed, Size, Cost and explain in detail. | 8 | L2 | 5 | 2 |
| b) Explain the role of virtual memories in computer systems with neat diagram of virtual memory organization. | 8 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
